

Cyberbullying Patterns in Twitter Data

SQL analysis of ~47,000 labelled tweets in Google BigQuery

The question: do harmful tweets attract more engagement than neutral ones? On this dataset they do not. The measurable separation between classes sits in tweet length.

~47K

Tweets analysed

6

Labelled classes

2020–23

Years covered

0.04%

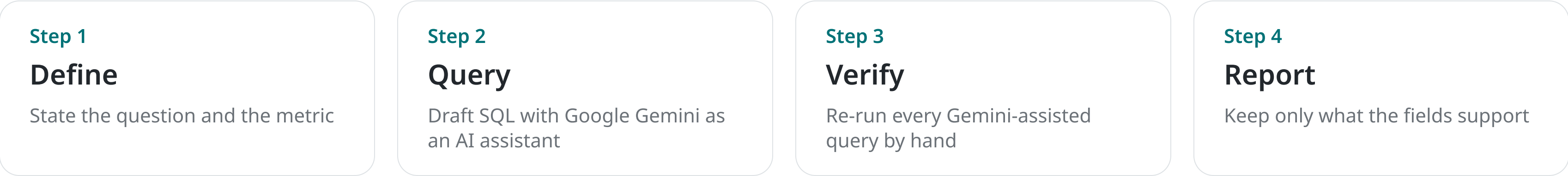
Rows with null dates

- 1 Engagement means are near-identical across all six classes (1.5% range).
- 2 The leading harm class rotates each year while annual volume stays flat.
- 3 Mean tweet length differs 2.4× between classes (197.9 vs 83.2 characters).

KEY TAKEAWAY Descriptive result: engagement did not separate the classes; tweet length did.

A balanced dataset and a repeatable query loop

Six labelled classes of roughly equal size, queried in BigQuery with standard SQL.



- Public Twitter corpus labelled by cyberbullying type, loaded into BigQuery.
- All figures are group means over the full corpus, not samples.

PROVENANCE CAVEAT
Engagement, date and location fields arrive with the dataset and could not be validated against the Twitter API.

CLASS BALANCE
Six labelled classes, each holding approximately 8,000 rows, for a corpus of about 47,000 tweets.

religion

age

ethnicity

gender

other_cyberbullying

not_cyberbullying

AI assistance: Google Gemini supported SQL drafting, query debugging and deck wording; all queries were re-run and all figures verified in BigQuery by the author.

KEY TAKEAWAY Equal class sizes mean cross-class comparisons are not weight-biased.

HYPOTHESIS

Do harmful tweets earn more engagement?

Amplification of harmful content is a common claim; these fields allow a direct descriptive test.

HYPOTHESIS — NOT SUPPORTED

Harmful tweets receive more likes, replies and shares than neutral ones.

- Metric: arithmetic mean of likes, replies and shares per class.
- Highest mean likes belong to not_cyberbullying (258.3), the neutral class.
- Highest-to-lowest gap on likes is 8.3 points, or 3.3%.
- No dispersion or significance testing was run, so this is descriptive only.

Class	Mean likes	Mean replies	Mean shares
not_cyberbullying	258.3	49.1	148.9
age	257.1	49.2	149.7
religion	256.1	49.5	151.0
gender	252.3	48.7	151.0
other_cyberbullying	251.9	49.5	148.7
ethnicity	250.0	49.3	150.3

KEY TAKEAWAY The direction of the hypothesis is not present in the data — a reportable result.

CROSS-CHECK

A combined metric reproduces the same result

Engagement score = mean likes + mean replies + mean shares, computed per class.

Engagement score — full range across all six classes



Range across classes: **7.0 points** on a base of ~450, or 1.5%.

SUPPORTED CLAIM

Within this dataset, mean engagement is effectively level across classes on both individual and combined metrics.

OUT OF SCOPE

No feed, ranking or exposure data is present, and engagement carries no timestamp. The result describes this static snapshot, not platform amplification.

KEY TAKEAWAY Two independent framings of the metric return the same conclusion.

The leading harm class rotates; annual volume does not

The top class holds close to 2,000 tweets in every year observed, but its identity changes.

Year	Leading class	Tweets
2020	religion	2,041
2021	religion	2,007
2022	ethnicity	2,042
2023	age	2,041

- Religion leads 2020 and 2021; ethnicity in 2022; age in 2023.
- Annual totals vary by 35 tweets across four years — under 2%.
- Possible driver, untestable here: shifts in public discourse.

DATA QUALITY CHECK

20 rows carried null dates and were excluded: 0.04% of the corpus, distributed across classes, so no single year or class is materially affected.

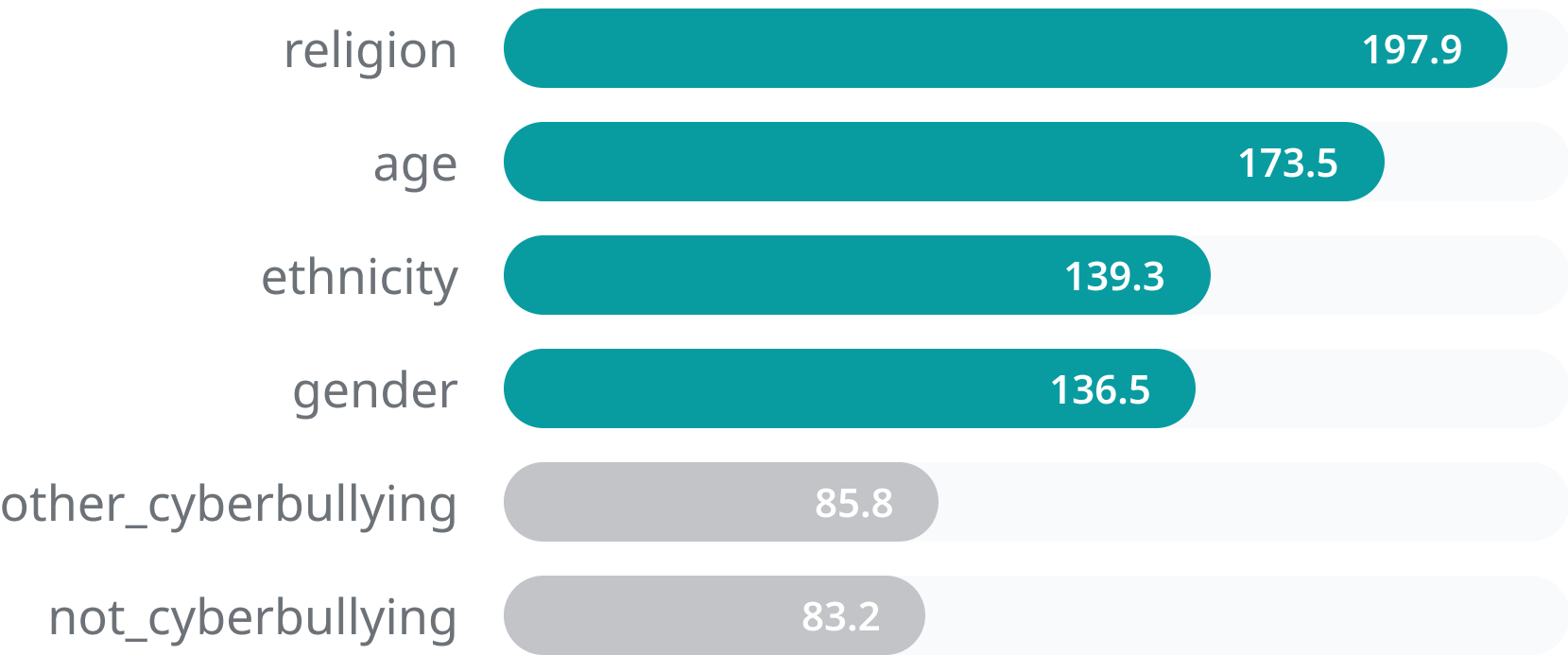
KEY TAKEAWAY Composition shifts year to year; overall volume stays stable.

PRINCIPAL FINDING

Mean tweet length differs 2.4× between classes

Religion averages 197.9 characters against 83.2 for not_cyberbullying — the widest separation found.

Mean tweet length by class (characters)



INTERPRETATION, UNTESTED

Longer texts may reflect argumentative abuse, shorter ones single-term abuse. Confirming that would require reading a labelled sample.

PRACTICAL IMPLICATION

Length is a cheap candidate feature for a classifier and is worth testing alongside text-based features.

KEY TAKEAWAY **The separating variable was length, not engagement.**

LIMITATIONS

What this dataset cannot answer

Location appeared informative until the collection method was examined.

Location	Most frequent class	Tweets
India	religion	1,634
USA	gender (tied)	1,199
Brazil	ethnicity	846
Canada	not_cyberbullying	842
UK	religion	833
Australia	gender	829

NOT A SUPPORTED RESULT

Location is self-reported free text with a high blank rate and no normalisation. The table is shown for transparency; no geographic conclusion is drawn.

Also absent: follower counts, exposure data, engagement timestamps, and any variance statistics. Every claim in this deck is descriptive and bounded by those gaps.

KEY TAKEAWAY Stating the boundary of a dataset is part of reporting the result.

A rejected hypothesis and a usable signal

Question

- Do harmful tweets out-engage neutral ones?
- Which measurable variable separates the classes?

Findings

- Engagement is level — 1.5% range across classes
- Leading harm class rotates yearly at flat volume
- Mean tweet length differs 2.4× between classes

Next steps

- Test length as a feature in a classifier
- Replicate on an independent labelled corpus
- Source data with verified timestamps and reach

The initial hypothesis was not supported, and the reportable result came from a variable that was not part of the original question: tweet length.